ACTIVITY AND MEASUREMENT TIMING

I do not want a timestamp mistake to look like model error.

My model predicts atoms. The experiments report activity measured after several real steps.

PERSONAL NOTE FOR

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My name is Samuel Ogunnubi, and I am a 12th grader in Maryland working on a reduced Ac-225 production model. I found NIST's Ac-225 standardization work while trying to compare my calculations with published experiments. That comparison looks simple at first because activity and atoms are connected by A = lambda N. The hard part is knowing which N and which time the reported activity actually represents.

THE FOUR CLOCKS I NEED TO KEEP SEPARATE

1

End of bombardment

Production stops and decay continues.

2

Chemical separation

Ra, Ac, and daughters may no longer share one history.

3

Assay time

The instrument measures activity under stated conditions.

4

Reported reference time

The published value may be decay-corrected to another time.

A = lambda N

The equation is not my main uncertainty. I need to know which decay constant and nuclear-data source to cite, whether progeny are in equilibrium, whether activity is for Ac-225 alone or a measurement window containing daughters, and how timing and uncertainty are carried through the conversion.

What I have already prepared

- Predicted atom inventories from the frozen reduced decay chain.
- Published produced and recovered activities with reported uncertainties.
- A timeline table for the Berkeley irradiation, separation, assay, and reference times.

What I need to get right

- The recommended decay data and activity-to-atoms procedure.
- How to state progeny equilibrium and the exact activity reference time.
- How measurement and nuclear-data uncertainty should appear beside model error.

My question: Could you recommend the best public method or reference for making this comparison correctly? I would use it to rebuild the external error calculation, separate measurement uncertainty from model error, and avoid claiming that an apparent mismatch came from V3 when it may have come from my timing or activity definition.

I would cite any guidance as a measurement reference only. I would not describe it as NIST validation of the production model.